



MBA 717M (Fall 2025)
Project Report

Geopolitical Risks and Governance in Rare Earth Mineral Value Chains

submitted by:
Md Waqar Moid (230644)
Anmol Agrahari (220161)

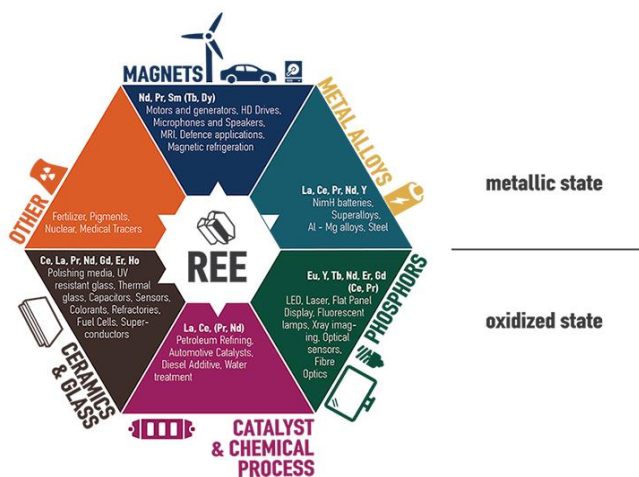
1. Introduction

Rare Earth Elements (REEs) represent one of the most critical, yet underappreciated, vulnerabilities in modern geopolitics. These seventeen minerals are basically the invisible foundation of 21st-century tech, powering everything from our smartphones to the advanced weapons systems that define military superiority. But the world is facing a major paradox: despite their name, REEs aren't actually that rare geologically. The crisis isn't about scarcity; it's about control [1]Through decades of patient, state-directed industrial strategy, the People's Republic of China has engineered a near-monopolistic position across the entire REE value chain. Right now, China controls about 70% of global mining, 91% of the critical refining capacity, and an incredible 94% of the high-performance magnets that drive electric vehicles (EVs) and wind turbines. [2] This concentration creates what analysts call a "single point of failure," threatening the energy transition goals and national security interests of nations worldwide.

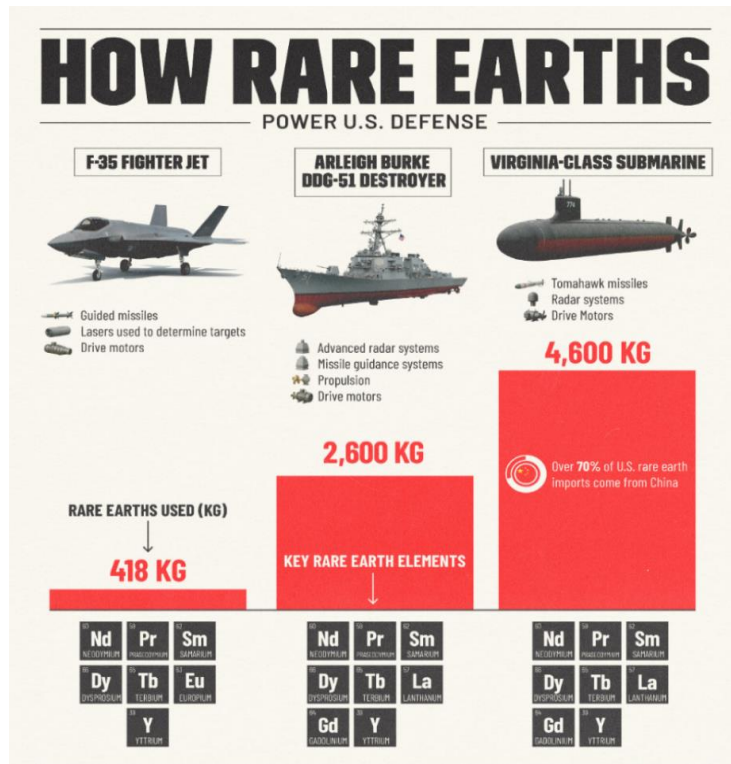
The Core Challenge - The main global risk isn't a lack of REEs in the ground, but the engineered geopolitical concentration of processing capacity. China's dominance forces Western nations to confront a fundamental market failure, where short-term economic efficiency (getting the cheapest materials) conflicts with long-term strategic resilience (not being dependent on a single supplier).

2. Understanding Rare Earth Elements

2.1 What Makes Them Essential?



REEs are prized for their unique magnetic, luminescent, and catalytic properties that just can't be easily replicated by other materials. [3] These characteristics make them indispensable for both civilian and military tech. In the civilian sector, they're the backbone of technologies driving the global energy transition, like EV motors and wind turbine generators.



The defence sector's dependence is just as profound. A single F-35 fighter jet, for example, requires 920 pounds of REE materials. [4] REE permanent magnets and special alloys are vital for jet engines, missile guidance systems, antimissile defence, satellites, and secure communications. The fact that some of these critical components are only made in China creates a direct national security vulnerability.

2.2 The Value Chain

To really get the REE challenge, you have to see that "scarcity" and "supply difficulty" are totally different problems. The bottleneck isn't finding deposits; it's in the hazardous, super expensive, and technically complex process of turning raw ore into usable products. [5]

- Upstream Stage (Mining Operations):** This is just digging the ore out of the ground. While China leads (240,000-ton quota in 2023), other countries have big operations too. The US produced around 43,000 tons in 2023, and Australia 18,000 tons. The "rarity" of REEs comes from them being scattered and mixed with other minerals, making it hard to find concentrations high enough to be economically viable without the processing infrastructure.
- Midstream Stage (Refining and Separation):** This is the critical chokepoint where China's dominance is most severe (91% control). Raw ore has to go through complex chemical and metallurgical processes to separate the 17 different REEs. This stage is incredibly capital-intensive, involves toxic chemicals, and requires specialized technical expertise. Because of this, ore mined in the US or Australia often just gets shipped to China for processing, which

means we surrender control over the whole supply chain anyway.

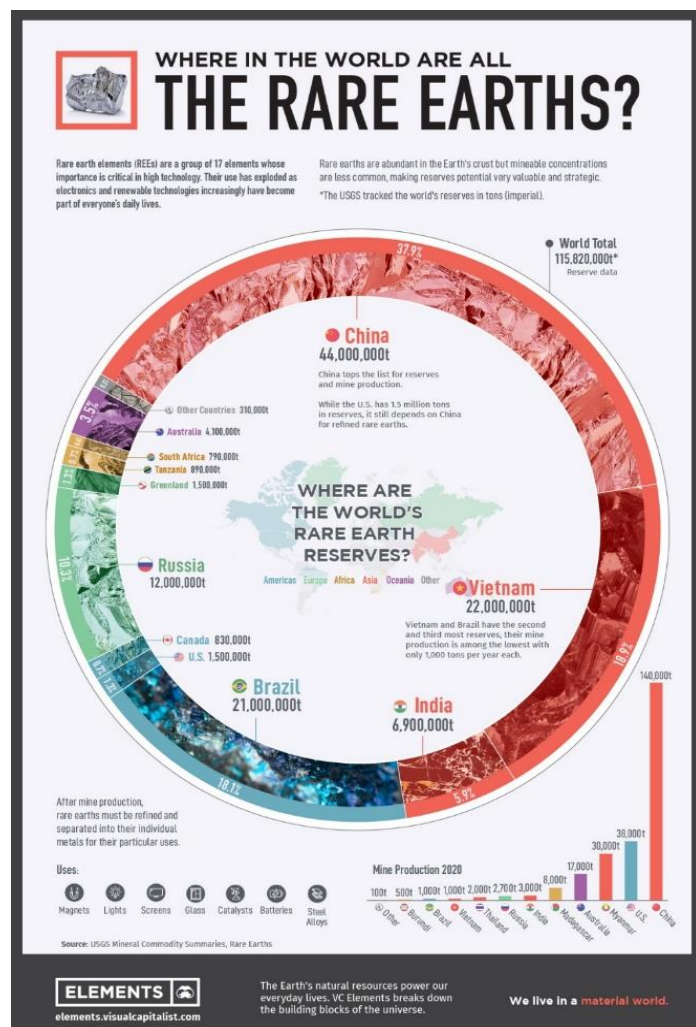
- **Downstream Stage (Manufacturing High-Value Products):** Here, the separated REE oxides are converted into metals, alloys, and most importantly, neodymium-iron-boron (NdFeB) permanent magnets. These magnets are the highest-value product and are essential for EV motors and wind turbines. China's 94% market share in magnets is a direct result of its control over the refined inputs from the midstream.
- **End-Use Integration:** This is the final stage where components are put into finished products, from Teslas to F-35s. Even Western companies end up depending on Chinese suppliers for these critical inputs.

3. Global Distribution and Historical Development

3.1 Geographic Distribution of Reserves

We look at where the REE reserves actually are, the picture is more diverse than the production numbers suggest. This is key to understanding if diversification is even possible.

The adjacent image clearly shows that China's production dominance (around 70% of mining) far exceeds its share of global reserves (37%). This discrepancy highlights how China's market position was built not just on geological luck, but on deliberate industrial policy. [6]



3.2 The Rise and Fall of American Leadership



The Mountain Pass rare earth mine and processing facility, owned by MP Materials, is viewed from an airplane flying at 37,000 feet on Sept. 1, 2023, in Mountain Pass, California. George Rose/Getty Images

It's quite surprising to think that the United States used to be the undisputed leader in REE production. From the mid-1960s through the 1980s, the Mountain Pass mine in California (owned by Molycorp) was the world's main source of REEs. American production literally powered the development of modern military tech, creating a cycle where defence needs drove demand, which supported domestic production. [7]

But this leadership position eroded. Decades of underinvestment from both government and industry, combined with losing specialized knowledge and skilled labour, weakened America's position. The final straw came in 2002 when environmental violations forced Mountain Pass to close. The US went from being self-sufficient to a net importer, with 80% of its REEs coming from China. [8] The stats are dramatic: in 2009, the US had 13% of global reserves; by 2019, that share had fallen to just 1%. Even though production has recently ticked up, the US still can't meet its own demand and, crucially, lacks the refining capacity to process its own ore. [9]

4. How China Engineered a Strategic Monopoly

China's dominance over the REE supply chain isn't a market accident. It represents one of the most

successful industrial policies of the last 50 years, executed through a patient, multi-decade strategy that turned geological resources into geopolitical leverage. [10]

4.1 Market Capture (1990s-2014)

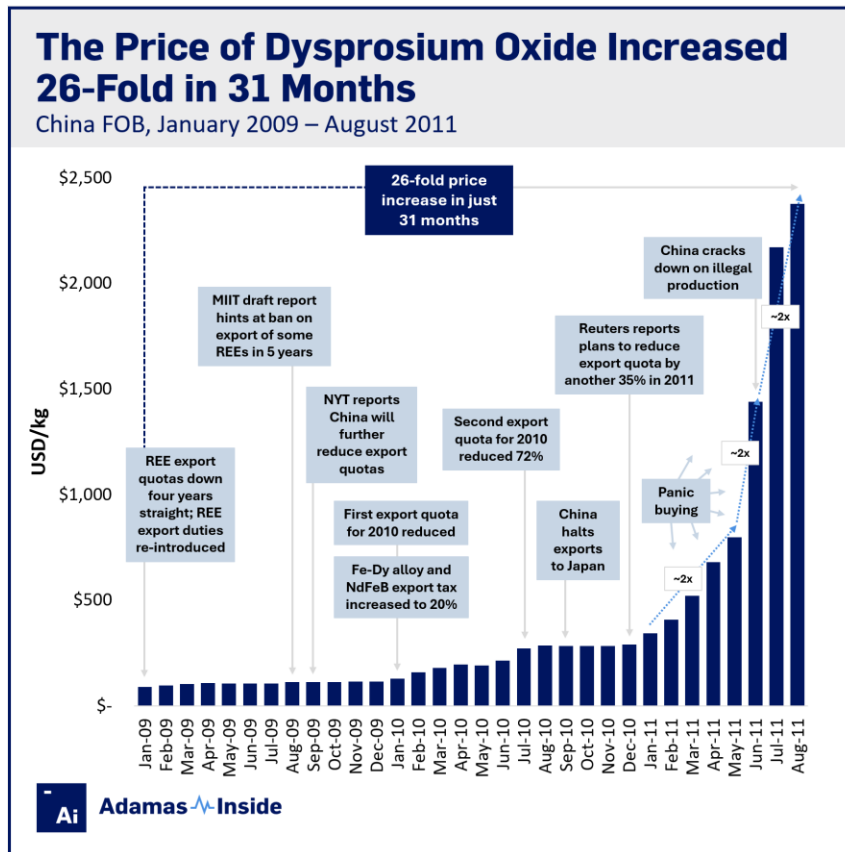
China's first move was to exploit an advantage Western competitors couldn't match: a willingness to absorb the massive environmental costs of REE processing. For decades, China invested in national programs and labs focused only on REE science. By the 80s, this investment paid off, positioning China to rapidly expand production. [11]

Chinese leaders labelled REEs as "strategic" minerals. Leveraging the country's tolerance for environmental degradation and low costs, they flooded the market in the 1990s. The environmental cost was staggering. Producing a single ton of REEs generates around 60,000 cubic meters of toxic gas, 200 cubic meters of acid-laced sewage, and over a ton of radioactive waste. [12] By "externalizing" these costs (i.e., not paying to safely manage them), China could sell REEs at prices far below what Western producers, operating under strict environmental laws, could match. This "environmental arbitrage" was brutally effective. Global REE prices crashed, forcing Western mining companies into bankruptcy. The Mountain Pass closure in 2002 was the symbolic end of this phase, as US producers just couldn't compete.



Pipes coming from a rare-earth smelting plant spew into a tailings dam on the outskirts of Baotou in China's Inner Mongolia autonomous region. Photograph: David Gray/Reuters

4.2 The 2010 Crisis: Weaponization of Supply



China's willingness to use its REE dominance as a weapon became terrifyingly clear in 2010. After a diplomatic incident involving a Chinese trawler and the Japanese coast guard, Beijing shut down all REE shipments to Japan for almost two months. [13] This disrupted Japan's auto industry and caused global prices for some REEs to spike over 1,000%.

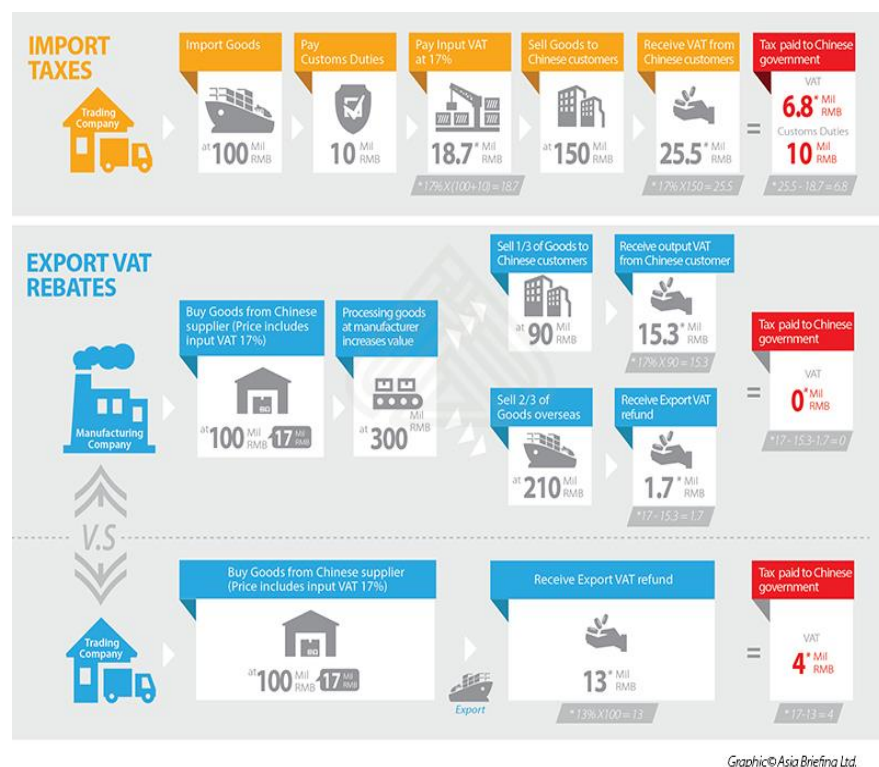
The 2010 crisis was a massive wake-up call for the

world. It proved China would use REEs as a geopolitical weapon, and it exposed the West's complete inability to fill the supply gap. It showed that while higher prices could in theory make old mines profitable again, restarting them takes years. Most importantly, it finally got Western governments to recognize their vulnerability, even if they didn't act effectively right away.

4.3 The Pivot to 'Smart' Control (Post-2015)

The 2010 crisis did backfire in one way. The US, EU, and Japan challenged China's export quotas at the World Trade Organization (WTO), and in 2014, the WTO ruled they were illegal. [14]. China's response, however, was incredibly sophisticated. Instead of retreating, Beijing pivoted to more subtle, WTO-compliant tools to achieve its real goal: capturing the entire high-value downstream industry. The main tool in this "smart" control strategy is the Value-Added Tax (VAT) rebate system.

The VAT Rebate Mechanism: This policy is pretty brilliant in its effectiveness. If a Chinese company exports raw REE oxides, it pays the full 13% VAT. But, if that same company uses those REEs to make a value-added product inside China (like a magnet), it gets a full 13% VAT rebate



when it exports the finished product. This simple tax rule creates an unbeatable 13% cost advantage for Chinese manufacturers. It makes building a magnet factory anywhere but China economically irrational. The policy effectively sucked the entire global magnet industry into China while starving potential foreign competitors of the refined REEs they'd need.

This was all backed by massive state subsidies (estimated \$9-10 billion between 2010-2019) and a consolidation of the domestic industry into six giant state-owned enterprises. China also created strategic stockpiles, giving it the power to manipulate global markets by releasing or withholding reserves. [15]

4.4 Consolidating Control: Recent Developments

In recent years, China has been even more explicit. In May 2020, President Xi Jinping visited a magnet factory. Days later, the state-run press warned that "by waging a trade war against China, the United States risks losing the supply of materials that are vital to sustaining its technological strength." [16] In June 2020, after a US missile component sale to Taiwan, Beijing threatened sanctions on US defence contractors and explored export controls specifically targeting the US and Europe. [17] This is a fundamental shift from market competition to the explicit weaponization

of a supply chain.

5. Allied Counter-Strategies: Building Resilience

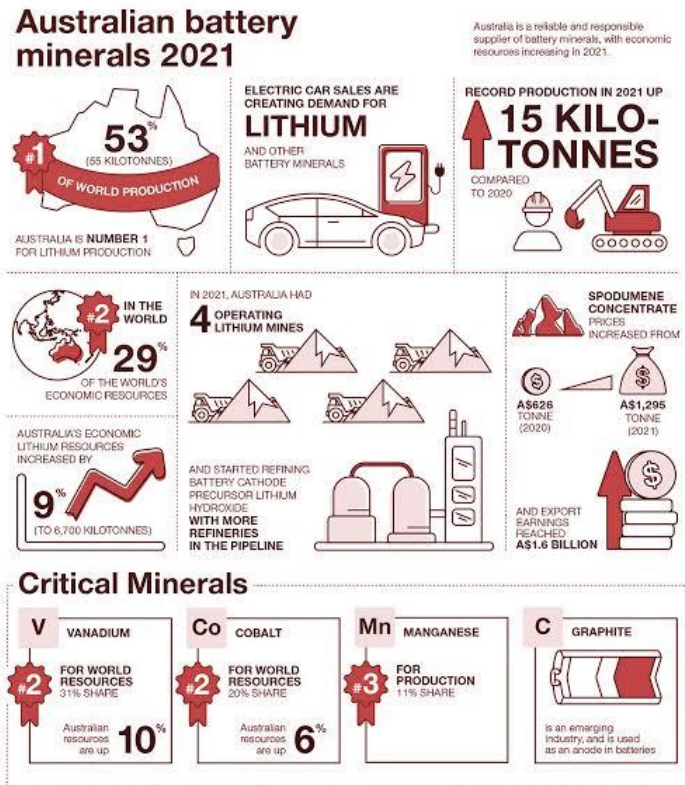
This recognition of vulnerability has finally translated into action, though it's a long road. Western nations are now pursuing a "China-Plus-One" supply chain strategy. This isn't a normal industrial policy; it's a national security project. Governments are having to actively create markets that the private sector won't build on its own. [18]

5.1 United States: The Defense Production Act Approach

US strategy is being led by the Department of Defense (DoD), which sees REE dependency as a direct threat to military readiness. Using the Defense Production Act, the DoD has committed over \$439 million since 2020 to build a "mine-to-magnet" supply chain completely separate from China. Key investments include a \$258 million contract for Lynas Rare Earths (an Australian firm) to build a heavy REE processing facility in Texas. This would give the US domestic refining capacity for the first time in decades. The effort, started under the Trump administration (EO 13817) [19] and continued under the Biden administration (EO 14017), shows a sustained political commitment.



5.2 Australia: Positioning as the Secure Western Supplier



Australia has positioned itself as the indispensable, secure supplier for the West. Lynas Rare Earths, the largest non-Chinese producer, operates a two-stage model: it mines ore at Mount Weld in Western Australia and processes it at a refinery in Malaysia. With US government backing, Lynas is now building more processing facilities in both Australia and the US, aiming to create a fully integrated Western supply chain.

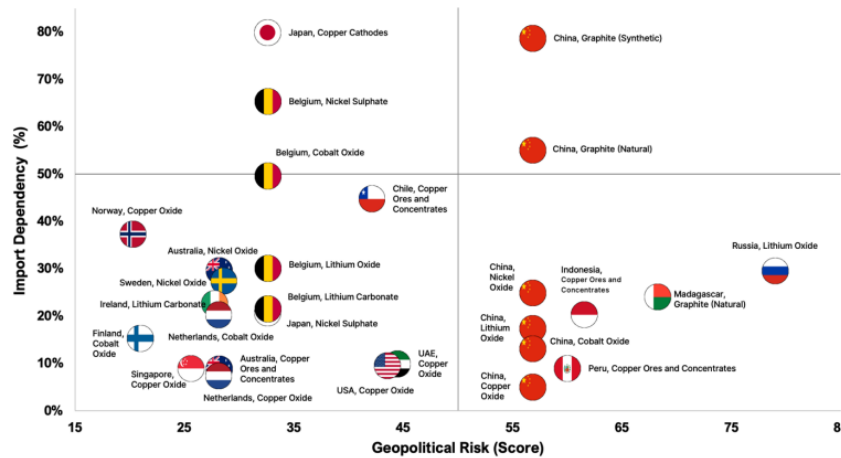
5.3 India: Pursuing Strategic Autonomy

India's approach is about "strategic autonomy"—reducing dependence on both China and the West. Led by the state-owned Indian Rare Earths Limited (IREL), India has an ambitious goal to triple its REE oxide output by 2032. The government is planning a subsidy scheme worth about \$600



million to build a domestic magnet industry, directly copying the Chinese model.

6. The Market Failure: Understanding the Economic Challenge



Source: UN Comtrade, Control Risks, Fragile States Index, IEEFA

The data reveals a fundamental market failure that explains why decades of warnings did nothing. The "free market" is structurally incapable of pricing long-term geopolitical risk into short-term purchasing decisions. [20]

6.1 The Two-Product Problem

Imagine you're an automaker sourcing magnets for your new EV. You have two choices:

- **Product One: The "China Price" Magnet:** This magnet is cheap, abundant, and available now. Its low price reflects state subsidies, the VAT rebate, externalized environmental costs, and economies of scale. If you're a manager judged on quarterly cost savings, this is the obvious choice.
- **Product Two: The "Security Premium" Magnet:** This is a theoretical magnet from a new US or Australian facility. It costs 30-50% more because it includes the true costs of environmental compliance, higher wages, and the billion-dollar investment to build the factory.

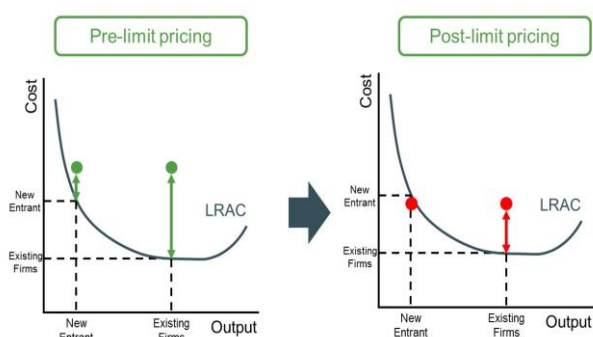
No rational CEO, whose job is to maximize shareholder returns, would voluntarily choose the "Security Premium" magnet. The market failed to create a secure supply chain because "efficiency" was defined only as "lowest cost," not "most resilient."

6.2 Why Government Intervention is Necessary

This explains why government action is absolutely necessary. China's VAT rebate is a state tool designed to tilt the market. Therefore, US Defense Production Act funding isn't "distorting" a free market; it's a counter-tool trying to create a new market for a product the private sector can't build

alone: a geopolitically de-risked magnet. The investment (over \$1 billion and 10+ years for one facility) is just too big and risky for private investors, especially when they remember Molycorp and other ventures that went bankrupt when China manipulated prices. [21]

7. Limit Pricing: China's Economic Weapon



"Limit pricing" is a key economic strategy China has used. A monopolist deliberately drops prices to a level where new competitors can't make a profit, or even enter the market. It lowers the monopolist's short-term profits but secures their long-term dominance. [22]

7.1 How Limit Pricing Works in Rare Earths

China has historically flooded the market with cheap REEs whenever it sees credible competition emerging.

- **Undermining Competitors:** When a Western mine plans to restart, China can depress global prices by increasing its own quotas or releasing reserves. This makes the new competitor economically unviable before it even gets started.
- **Discouraging Recycling:** Recycling e-waste ("urban mining") is a potential alternative supply. But it has high setup costs. When China floods the market with cheap primary REEs, the economics of recycling collapse, discouraging investment.
- **Eroding Stockpile Incentives:** Governments could build strategic stockpiles (like strategic petroleum reserves). But it's hard to justify spending billions on a stockpile when prices are artificially low, and there's a risk the tech could change, making your stockpile obsolete.

7.2 Limits to China's Pricing Power

While this strategy was effective, China may not be able to maintain it indefinitely. First, China's own domestic demand for REEs is surging as its clean energy sector booms. Domestic demand now actually exceeds domestic supply for some REEs. [23] Second, using limit pricing would

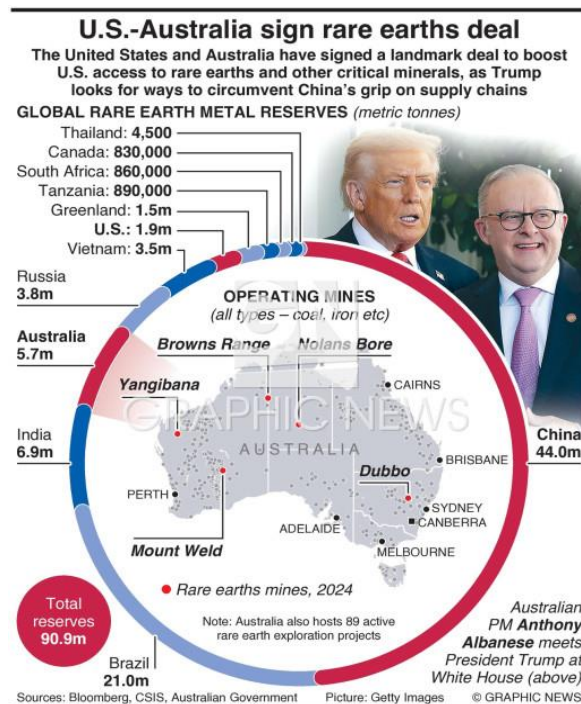
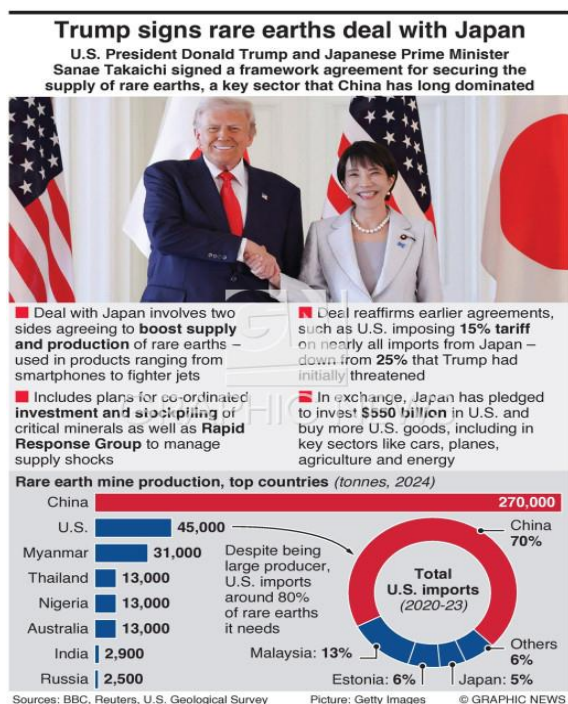
mean diverting valuable refined REEs away from its own high-value manufacturing industries, which are now major drivers of the Chinese economy. These constraints create a window of opportunity for Western producers, if they can get enough government backing to survive short-term price manipulation.

8. Comprehensive Policy Recommendations

The era of a single, globalized REE value chain is over. The world is splitting into two parallel systems: a low-cost, state-controlled Chinese chain and a high-cost, state-supported allied chain. Making this allied chain viable requires a coordinated strategy, building on frameworks from global value chain governance theory. [24]

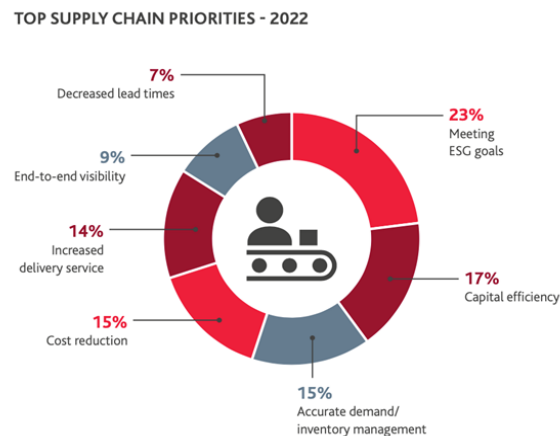
8.1 Form a 'Buyers' Club' with Guaranteed Procurement

Allied governments (US, EU, Japan, Australia, India) must create a formal "Minerals Security Partnership." This group must pool its demand and sign long-term, guaranteed purchase agreements (offtake agreements) for new REE projects. This is the only way to "de-risk" the billion-dollar-plus private investments needed.



8.2 Compete on Transparency, Not Price

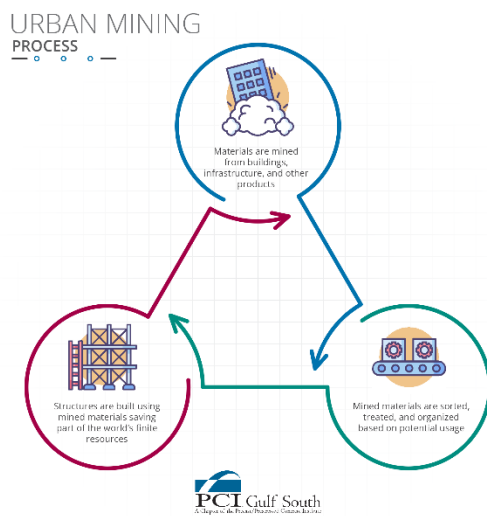
Allied nations can't beat China on price. The solution is to change the game by creating a premium market for "certified clean and secure" REEs. By using verifiable certification (maybe on a blockchain), Western producers can offer something China can't: supply chain transparency and environmental accountability. This gives ESG-conscious companies (like automakers) a legal and public justification for paying a premium.



2022 BDO Technology CFO Outlook Survey [34]

8.3 Invest in Circular Economy and Recycling

The most secure REE is one that's recycled. Substantial investment in "urban mining" to reclaim REEs from end-of-life products (e-waste, old batteries, etc.) creates a domestic supply source that is immune to geopolitics. [25] Governments should fund R&D for better recycling processes and provide incentives for companies that use recycled content.

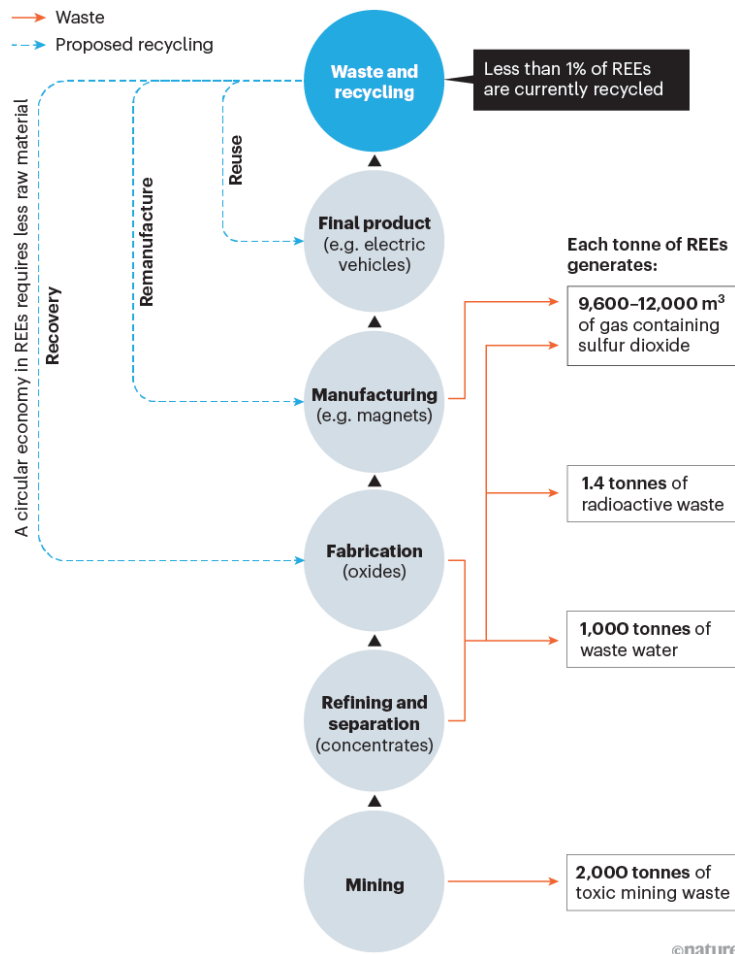


8.4 Build and Maintain Strategic Stockpiles

While new production ramps up, nations need an

A CIRCULAR ECONOMY IN RARE EARTHS

If products containing rare-earth elements (REEs) were recycled, secondary markets could supply half of industry's needs. This would also reduce the environmental impacts of mining and refining.



immediate buffer. The US National Defense Stockpile needs to be expanded to sustain critical manufacturing for at least 3-5 years, and it should include processed oxides and finished magnets, not just raw ore. [26]

8.5 Develop Substitute Materials and Reduce Intensity

Technological innovation is another path. We need more R&D (like from DARPA) into substitute materials and new magnet designs that require less REE content. A breakthrough here could be a game-changer, similar to how hydraulic fracturing transformed US energy. [27]

8.6 Create Integrated Domestic Processing Industries

This is the most critical long-term goal. We must develop fully integrated domestic industries, from mine to refinery to magnet factory. Just mining more ore and shipping it to China for processing doesn't solve the problem. [28] This requires massive public investment in rebuilding the skilled labour force (university programs, vocational training) and streamlining the 10+ year permitting process for new facilities.

8.7 Enhance International Collaboration

No single Western nation can match China's integrated chain alone. Success requires

unprecedented cooperation. Allies must have reciprocal defence procurement agreements, security of supply arrangements (to ensure materials flow freely among allies during a crisis), and joint R&D initiatives. [29]

9. The Broader Economic and Environmental Context

9.1 The Electric Vehicle Challenge

Hybrid and electric cars can contain 20–25 pounds¹ of rare earths
(Twice the amount found in standard gasoline cars)



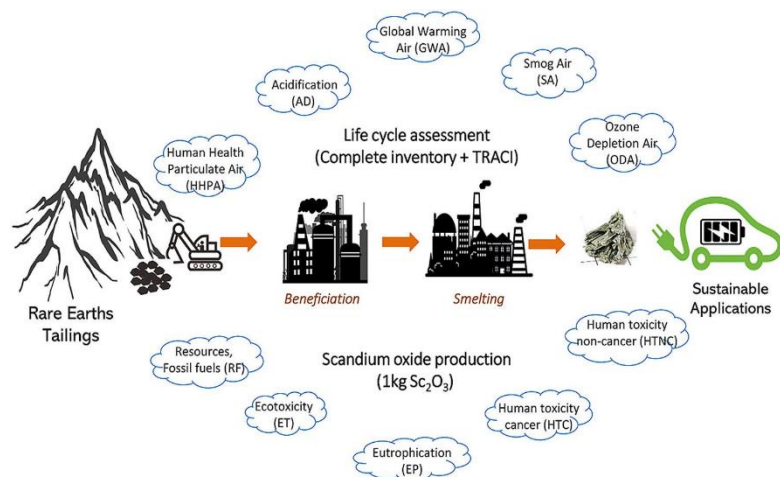
¹Source: "The Race for Rare Metals", Globe and Mail, July 16, 2011

The REE vulnerability goes way beyond the military; it is at the heart of the global energy transition. The auto industry's commitment to EVs is creating a massive surge in demand for REE magnets. Right now, China has over 70% of global EV

battery manufacturing capacity (the US has less than 10%). [30] Beijing has explicit goals to produce 50% of the world's EVs by 2025. The scary result could be a world where Western nations hit their climate goals but become even more dependent on China for the tech that got them there.

9.2 Environmental Considerations

This whole issue presents a real dilemma. China's market capture strategy was built on externalizing environmental costs (60,000 cubic meters of toxic gas per ton of REE, etc.). [31] That model is not sustainable or acceptable for Western nations. The challenge is to find ways to make REE



production environmentally responsible without making it economically impossible. This requires new, cleaner processing tech and transparent accounting of environmental costs. The "premium price" strategy for certified-clean REEs only works if Western producers can actually prove they are substantially better for the environment.

10. Conclusion: A New Paradigm for Strategic Resources

The REE crisis is more than just a supply chain problem. It highlights a fundamental tension in how modern economies balance efficiency, security, and resilience. For decades, the main economic doctrine was that markets should optimize for efficiency, period. The REE situation proves the profound limits of that model when dealing with critical resources that can be weaponized.

China's patient, multi-decade strategy worked because it understood something the West forgot: control over critical inputs gives you geopolitical leverage that is more valuable than just economic value. The West's response has been slow, but a window of opportunity is opening. China's own resource constraints and surging domestic demand limit its ability to use limit pricing as effectively as it did in the past. [32] This creates space for alternative supply chains, but only if they get enough government support to get off the ground.

Success requires us to recognize that this is a governance challenge, not just a mining problem. Market forces alone will not create the supply chain diversity that national security requires. Government intervention, through procurement guarantees, subsidies, and international coordination, isn't a "distortion" of the market; it's a correction of a massive market failure.

The policy recommendations outlined here—a "Buyers' Club," transparency, recycling, stockpiles, R&D, and allied cooperation—are a comprehensive strategy. But they require sustained political will and tens of billions of dollars over decades, not years. The alternative is to remain dependent on a strategic competitor that has already shown it's willing to weaponize that dependence. As the world moves toward green tech, the nations that control REE supply chains will have disproportionate influence over the economy, the energy transition, and military power.

The rare earth challenge is a template for other critical supply chains (semiconductors, pharma, batteries). Establishing a resilient REE supply chain requires us to abandon the idea that economic

interdependence always leads to peace. In a world of great power competition, controlling critical resources is a form of power that has to be balanced.

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